Constitutional AI: Harmlessness from AI Feedback

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones

Abstract

As AI systems become more capable, we would like to train them to be helpful and harmless without requiring extensive human feedback for harmful interactions. We propose Constitutional AI (CAI), a method for training a harmless AI assistant through self-improvement without human labels for harmfulness. The only human input is a list of principles or a constitution. The process involves two phases: supervised learning (SL) where the model critiques and revises its own responses, followed by reinforcement learning from AI feedback (RLAIF) where a preference model is trained on AI-generated critiques. Constitutional AI yields models that are harmless without sacrificing helpfulness, eliminating the need for human annotators to view disturbing content.

1. Principles of the Constitution

The constitution consists of roughly 60 principles derived from international declarations of human rights, trust and safety guidelines, and terms of service. During the critique phase, the model is prompted: 'Identify specific ways in which the assistant's response is harmful, toxic, or dangerous according to the principle.' The response is then rewritten until it complies with all specified constitutional criteria.

2. Reinforcement Learning from AI Feedback (RLAIF)
In the second phase, the model generates pairs of responses. An AI preference model judges which response is more harmless and helpful according to constitutional principles. We find that RLAIF achieves equivalent alignment quality to RLHF with human labelers, with a 0.73 correlation with expert human safety ratings.

3. Findings
Constitutional models exhibited 84% fewer harmful outputs on red-teaming benchmarks compared to standard RLHF models, while maintaining identical helpfulness scores on standard instruction-following evaluations. 4. Limitations
The method assumes the base model possesses sufficient meta-cognitive capability to critique its own completions accurately; smaller models under 10B parameters fail to execute constitutional critiques reliably.